SL1680 Edge-AI Terminal · Project Overview

2026-07-28 · v2.7 · Synaptics SL1680 (Astra Dolphin RDK) · the single top-level entry

Local Voice AI | DL7400 4-Display | TD7800 LVDS+Touch Direct Drive | Power-cycle recovery 9/9

1. The Project in One Sentence

An **edge-AI human-machine terminal** on the Synaptics SL1680 (4×Cortex-A73 + NPU): local voice AI (ASR/TTS fully on-board) + two display chains (DisplayLink DL7400 USB multi-display + TC358775 direct-drive LVDS touch panel) + visual wake-up — all self-recovering after power loss.

Test conditions: open-source Astra SDK from GitHub (Yocto scarthgap) + one borrowed SL1680 RDK and DL7400 EVK + self-purchased peripherals (C920 camera, 4 self-owned monitors, TM10.5-TD7800 touch panel, LAMTL adapter, amp/speaker).

2. System Architecture

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[C920 camera]-USB-> NPU: person detect 25ms/frame -> visual wake greeting
[C920 mic]   -USB-> CPU: silero VAD -> SenseVoice ASR(int8) -> 4-layer routing
   (1)local volume (2)local device (3)web search+grounding (4)cloud DeepSeek
   -> matcha TTS -> ALSA astraout -> [HT517 amp + speaker] (I2S)
Display A: dl_face -> GStreamer -> [DL7400] -USB3-> up to 4 monitors
           screen0 = face + bilingual subtitles, screen1 = live camera
Display B: weston -> DSI 4-lane -> [LAMTL/TC358775] -LVDS->
           [TM10.5-TD7800 1280x720 touch panel] (touch back ~120Hz)
Cloud: DeepSeek + Zhipu standalone search | Offline fallback: local qwen2.5-0.5b
```

3. Three Achievement Domains

3A. Local Voice-AI Terminal (v2.0-v2.6)

Capability	Key points	Measured
Fully local speech chain	SenseVoice ASR (int8) + matcha TTS, no PC	ASR RTF 0.197; TTS 0.37
4-layer routing	Local commands really execute; realtime Qs use real search + grounding; chat to DeepSeek	"Volume 20%" -> amixer [20%]
Visual wake-up	NPU person detection -> greeting	25 ms/frame
Microphone guard	Card-name + capture-node wait; survives 86 s USB flap	Adversarial: 0 model reloads
Stability	Full power-cycle recovery; daily cleanup	Re-verified

3B. DL7400 USB Multi-Display

Conclusion	Data
4 simultaneous monitors feasible	8.7 / 5.8 / 15.0 / 5.7 fps concurrent
USB3 total bandwidth constant	3 to 4 screens: 4K drops 7.2 to 5.8 fps (-20%)
Bottleneck is DL streaming	111 -> 31.9 (HW scale) -> 7.4 fps stage-by-stage
3 SDK defects reported to DisplayLink	hotplug callback / EDID cache / dpaux blackout
No self-heal after monitor power loss	Workaround: voice "re-detect screens"

3C. TD7800 LVDS Touch Panel Direct Drive (v2.7, six root causes peeled)

#	Root cause	Fix
1	Raspberry-Pi panel leftovers in base device tree	Replace dsi_panel with TM10.5 timing

#	Root cause	Fix
2	TC358775 powers up tri-stated, uninitialized	29 DSI generic writes baked into DTB
3	Panel RESX floating (whole TDDI in reset)	Tie high 3.3V in hardware
4	U-Boot muxed SPI data pads to UART2 flow control	Reclaim via dts pinmux (bonus SPI debug channel)
5	Zero DSI-to-LVDS bandwidth margin -> live noise	Byte_clk 65300 + PCLKDIV=4 (+33% margin)
6	Touch INT GPIO bank ambiguous	Located empirically: porta10

Companions: SPI debug tool (ID/BIST/NVM read), 120 Hz multitouch, 180-degree rotation + touch calibration, Shanghai timezone. Power-cycle recovery 9/9 measured.

4. Key Knowledge Distilled (reusable)

1. **DSI-to-LVDS bridges need bandwidth margin**: LVDS pclk = DSI clock-lane freq / PCLKDIV; zero margin = line-buffer underrun = live noise
2. **TDDI reset discipline**: TP_RST/RESX always tied high in hardware - floating kills everything; SoC control kills the display
3. **Level-domain discipline**: 3.3V into a 1.8V pin back-feeds via ESD diodes (measured 2.38V)
4. **Never trust EDID preferred_mode**: a 160 Hz preferred mode blocked DL7400 forever; the fault travels with the monitor
5. **Non-48 kHz rates crash the audio driver**: the astraout resample chain is a necessity; address ALSA by CARD=name
6. **Zhipu chat's web_search is fake**: use the standalone search API + grounding; answer "cannot find" rather than fall back
7. **Four Yocto trap classes**: FetchContent hard-off (mimics network failure) / unversioned .so mis-packaged / :pn- only in conf files / build-green != feature-complete
8. **The board is the truth**: backflow immediately, reconcile sha256
9. **Three laws of remote ops**: script files for complex commands; check busybox gaps; guard binaries against BOM

5. Software Assets

Asset	Location	Notes
Yocto layer	meta-dlsdk/ (VERSION v2.7)	astra-voice / dl-face / sherpa-onnx / dlsdk / linux-syna.bbappend (TD7800 patch)
Full image	flash_image_20260723163055/	SYNAIMG 2.5 GB, flash-and-go (models separate)
TD7800 boot bundle	SL1680_td7800_boot/	3 image generations + scripts + retrospective
Model backup	backups/models_2026-07-24/	1.15 GB sole copy (contains plaintext keys)
Debug-code backup	TD7800_bringup_backup_2026-07-28/	79 files + SHA256
Rootfs trio	/etc on board	rotation / touch calibration / timezone (to Yocto)

Build: bitbake astra-media; kernel-only rebuild 75 s; boot dd both slots + read-back verify, ~4 min end-to-end.

6. Hardware Assets and Wiring

Device	Interfaces	Wiring doc
SL1680 RDK	J208 (DSI 22P) / J32 (40P header)	Wiring Reference sec.1/3
LAMTL adapter	J1 (DSI in) / J3 (LVDS out) / J2, J4 unused	ibid. sec.1/2
TM10.5-TD7800	U4 (LVDS+touch) / U5 (power/SPI/reset) / U3 (backlight)	ibid. sec.2/3
DL7400 EVK	USB3 upstream + 4x DP/HDMI	plug & play
Audio kit	HT517 (I2S/J32) + speaker; mic = C920	Xiaozhi-kit guide
External supplies	12V backlight (via driver board) / 3.3V panel+resets / 1.8V LAMTL	Wiring Reference sec.4

7. Current Running State (measured 2026-07-28)

Item	State
Auto-start services	astra-voice / astra-translate / dl-face / vision-wake / cleanup.timer ON; astra-mode, astra-xiaozhi, dl-clock disabled

Item	State
Displays	DL7400 multi-screen OK + TM10.5 LVDS desktop OK (180-degree, CST clock)
Touch	0x2C claimed, IRQ porta10, evtest reporting, calibrated
boot	boot_td7800_touch.subimg (sha 0aa90172...) both slots; rollback backups on board
Power-cycle recovery	9/9 self-check passed

8. Known Open Issues (merged)

#	Issue	Priority
1	KWS wake word not integrated - false triggers burn cloud API; assets ready, only software integration missing	Highest
2	No AEC echo cancellation	High
3	DL7400 no self-heal (reported to DisplayLink)	Medium (vendor)
4	On-board mic ZTS6672 driver issue (FAE)	Medium
5	TTS licensing (espeak-ng GPLv3)	Medium
6	New image not flash-verified on real hardware	Medium
7	LVDS jumpers to twisted pairs	Low
8	Cleanall verification of TD7800 patch	Low
9	Rootfs trio into Yocto	Low

9. Document Navigator

Top level: this document (CN/EN, md+html+pdf) -> SL1680 Project Summary 2026-07-23 (CN/EN, the v2.6 deep dive)

TD7800 panel topic: Display Bring-up Summary (CN/EN) -> Hardware Wiring Reference (CN/EN) -> TC358775 Config & Debug Record -> TD7800 SPI Source Code -> backup README (79-file inventory)

Reporting & comparisons: SL1680+DL7400 Achievement Report (CN/EN) - SL1680 vs ESP32 - robotics surveys - iCub paper digest

Operations: Backup & Recovery Manual - SL1680_td7800_boot/README (flash & rollback) - meta-dlsdk/VERSION - backups/CHANGELOG.md

10. Roadmap

1. **Integrate KWS** (kills false triggering and unlocks pause-by-voice)
2. Real-hardware flash verification + rootfs trio into Yocto -> a one-flash full-function deliverable
3. Robotics: SL1680 torso compute + TM10.5 face + DL7400 teleop console + finger actuators - aligned with Synaptics' high-speed-interfaces-for-robotics narrative